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Zixu Wang

PhD Student • Lancaster University

Profile

Full-time PhD student at Lancaster University researching quantum-inspired agentic AI for financial modelling. Python and PyTorch experience spans data preparation, model training, evaluation and explainability across medical imaging, ECG and LLM projects. First author of a PRCV 2026 paper, seeking AI/ML engineering opportunities.

Education

Lancaster University — Lancaster, UK
PhD (full-time) — **Oct 2026 – Present**
Supervisor: Dr. Richard Jiang. Research: quantum-inspired neural agentic AI for autonomous financial modelling, focusing on tensor networks, reinforcement learning and explainable decision-making.

Lancaster University — Lancaster, UK
MSc Artificial Intelligence — **Oct 2025 – Sep 2026**
Expected Distinction; formal award pending. Dissertation: 75/100.

Aston University — Birmingham, UK
BSc Computer Science (Hons) — **Sep 2021 – Jul 2024**
Second Class Honours, Lower Division. Selected modules: Mathematics (79.0), Human–Computer Interaction (70.0).

Publication

Zixu Wang, Richard Jiang. **How Far Can a Single Lead Go? A Construct-Validity Audit of ICD-Based ECG Benchmarks on 54,357 MIMIC-IV Patients.** *PRCV 2026.*

- Benchmarked six model families with patient-level splits; audited ICD-derived labels using no-ECG baselines, age stratification, label sensitivity and zero-shot transfer to PTB-XL.
- Identified demographic and care-pathway contributions to aggregate AUC (no-ECG: 0.816; best ECG: 0.800), with ECG gains of 0.05–0.07 within age strata 50–79.

Research

MSc Dissertation • Region-Level Explainability on PD-DBS Facial Images — *Jun – Sep 2026*
Explainable Face2Gene Diagnosis project, Lancaster University | Supervisor: Dr. Richard Jiang

- Designed an eight-region occlusion audit and Anatomical Evidence Vector (AEV) pipeline on 2,343 grayscale 32×32 images, quantifying model sensitivity for pre-/post-DBS classification.
- Evaluated NumPy MLP and compact CNN models with calibration, similarity filtering, spatial permutation, region-only tests, atlas geometry variants and Grad-CAM comparisons.
- Four regional contrasts persisted after confidence matching and FDR correction; right cheek/zygomatic ranked first by balanced accuracy in 10 of 11 atlas variants.

Neglected Tropical Disease Clinical Trials Analysis — *Sep 2025 – Dec 2025*
SCC.450 Introduction to AI and Data Science | Python • Regression • Network Analysis

- Led a 2–3 member team analysing 315 WHO ICTRP trial records across 62 countries; cleaned data and built logistic-regression models, sponsor–country–disease networks and visualisations of clinical-research distribution.

Projects

Image Classification & Clustering — *PyTorch • DINOv2 • Clustering*
SCC.451 Machine Learning

- Applied transfer learning and fine-tuning with ResNet50, DenseNet121 and DINOv2 for image classification; used K-means, hierarchical clustering and PCA for climate-data analysis.

Diversity Beats Strength? Multi-LLM Teams in Game-Theoretic Settings — *LLM Agents • Simulation*
SCC.452 Intelligent Agents and Autonomous Systems

- Led a 4–5 person team and ran approx. 6,400 trials across three games using GPT-4o, DeepSeek-V3 and Gemini-3-Flash; evaluated cooperation, payoff, communication, memory, topology and team composition.

Sentiment Analysis across Multilingual Datasets — *NLP • English/Chinese*
SCC.453 NLP and Language Models

- Compared tokenisation, stemming and lemmatisation strategies for English and Chinese sentiment-classification pipelines, evaluating preprocessing impact on accuracy.

Undergraduate Projects: Restaurant Back-Office System & PC Components E-Commerce Site — *Aston University*
Individual / Team

- Built a restaurant backend/admin management system (final-year individual project) and a team-based e-commerce site for PC components, contributing to requirements, implementation, testing and delivery.

Technical Skills

- **Programming:** Python, SQL, Java
- **AI / ML:** PyTorch, scikit-learn, NumPy MLPs, CNNs, transfer learning, clustering, ECG analysis
- **NLP / LLMs:** Text preprocessing, OpenAI / Anthropic / Google API integration, agent simulations
- **Data / XAI:** pandas, NumPy, matplotlib, seaborn, NetworkX, occlusion, Grad-CAM, calibration, AUROC/AUPRC, FDR
- **Tools & Languages:** Git, Linux, Jupyter, LaTeX; Chinese (native), English (fluent)